

One bucket record measures both scene and medium

[author block — USER DECISION (gap G7)]

Abstract

[ABSTRACT — 80–100 words, DEFINE/DECODE/CERTIFY in miniature: (1) a bucket record is generated by a hidden dynamic channel usually corrected as nuisance; (2) a joint missing-information partition defines two ledgers of one hidden-state experiment and canonical-confusion reciprocity says when both stay identifiable; (3) one unchanged bucket record is decoded into scene and medium (t_c , CV) with no pilot or reference arm; (4) a preregistered campaign certifies medium precision, calibrated coverage, and scene noninferiority within a declared scalar-gain domain. Only numeric call-out permitted: $\leq 1.5 \times \text{pilot}$ ($\langle C:\text{pilot ratio} \rangle$). No M1/ridge numbers (binding cut). State simulation-only. Pre-verdict wording “two inferred products”; “model-certified” only after C1–C7 pass.]

1 One record, two products

A bucket camera records one number per pattern, but that number is generated by a hidden dynamic channel.^[GI/SPI core] The channel is usually treated only as distortion; here we ask whether the same imprint can serve as a medium measurement.

► TODO: Finish para 1 (1–2 sentences). Optionally place the ONE-sentence DCS distinction here (else para 3): diffusing-correlation/wave spectroscopy measures medium dynamics from a dedicated intensity stream, whereas this paper asks whether a programmable bucket stream carrying an *unknown scene* can deliver *both* products without a dedicated assay channel (R36 s1). Source: R36 s1 para 1; materials map s4 (DLGI).

Before asking whether the medium can become a second output, we tested whether ordinary bucket imaging still had meaningful scene-recovery headroom. Across five preregistered comparisons, changing the pattern family, redistributing measurements, reordering the same patterns, replacing the reconstruction algorithm, and using the full count statistics produced no material improvement over the global operating rule; one change was strongly harmful. The design-side searches and the inference-side tests therefore bracketed the same conclusion from opposite directions: within the declared count-only experiment, the remaining scene gain is below the threshold fixed in advance. This does not prove

optimality over every possible optical measurement, but it changes the next question from how to squeeze another fraction of a decibel from the bucket count to what else the hidden channel has already measured.

► TODO: Para 3 (claim + evidence). State: the two-ledger theorem; the unchanged acquisition ledger; pilot/oracle baselines; calibrated intervals; scene noninferiority; validity map. Keep it to a few sentences (no contribution list). Source: R36 s1 para 3. END the paragraph – and the section – with the frozen manuscript-contract sentence:

The hidden channel that limits bucket imaging is itself an instrument: one unchanged bucket record can reconstruct the scene and measure the medium, while canonical-confusion reciprocity states exactly when the two products remain simultaneously identifiable.

We first quantify the information removed by the hidden state before asking how that boundary should be used.

2 Two ledgers of one hidden-state experiment

► TODO: Opening sentence: credit the Louis / Orchard–Woodbury observed-information identity immediately; state that the contribution is its programmable-ghost-imaging specialization and the canonical-confusion reciprocity consequence. Source: R36 s2; DOI fence materials map s8c (Louis 10.1111/j.2517-6161.1982...).

$$p(y, a \mid x, \theta) = p(y \mid x, a) p_\theta(a). \quad (1)$$

► TODO: One sentence reading Eq. (1): the pattern/medium path a is generated by medium hyperparameters $\theta = (\log t_c, \log \text{CV})$ and couples to the scene x only through the bucket likelihood.

$$\mathcal{I}_Y(x, \theta) = \text{diag}(C_x, C_\theta) - \mathbb{E} \text{Cov}([U_x; U_\theta] \mid Y). \quad (2)$$

► TODO: One sentence: C_x, C_θ are the complete-data informations; the posterior score covariance is the price of observing only Y . Import R33 Thm 1 (materials map s4b): $\mathcal{I}_Y = \text{diag}(C_x, C_\theta) - L$, $L \geq 0$.

FIG. 1 (HERO) PLACEHOLDER --- the hidden channel becomes the instrument. Three visually continuous zones (not a dashboard). LEFT (one unchanged acquisition): patterns $\rightarrow$ scene $x \rightarrow$ dynamic medium $a_t \rightarrow$ bucket stream Y ; one detector, one time series; footer “same photons · same exposures · no pilot · no reference arm”. CENTER (the limit): one small posterior-ambiguity cloud around a_t ; “hidden-state uncertainty subtracts scene information”; ≤ 1 compact equation; NO ridge / map / campaign numbers. RIGHT (the reuse): the SAME raw Y bifurcates into scene $\hat{x}$ and medium $(\hat{t}_c, \hat{CV})$, joined by a thin arc labeled “shared canonical confusion” (NOT “conservation”/“transfer”); conclusion “one record $\rightarrow$ two inferred products” (“model-certified” only post C1--C7).

Figure 1: **[Fig. 1 title: the hidden channel becomes the instrument.]** [Caption: one unchanged optical record; hidden-state uncertainty as the scene limit; the same record decoded into two products joined by shared canonical confusion. No M1/ridge numbers (binding cut). No provenance.]

$$J_x = A - CB^{-1}C^\top, \quad J_\theta = B - C^\top A^{-1}C, \quad (3)$$

► TODO: Name J_x, J_θ the scene and medium efficient (oracle) ledgers, with A, B, C the blocks of Eq. (2). Import R33 (materials map s4b).

$$K = A^{-1/2}CB^{-1/2}, \quad \frac{\det J_x}{\det A} = \prod_i (1 - \kappa_i^2) = \frac{\det J_\theta}{\det B}, \quad (4)$$

where κ_i are the singular values of K .

► TODO: One-line reciprocity statement: both normalized ledger losses share the same nonzero eigenvalues κ_i^2 . Then the O4-A consequence (R36 s2 item 5), ONE sentence: $C = 0$ (equivalently $K = 0$) protects BOTH ledgers, not one at the expense of the other. FORBIDDEN wording (register B3): no “information conservation”, no “free medium information”, no “paired trades scene into medium”. Correct: “two projections of the same posterior ambiguity, coupled through the posterior cross-covariance K ”. Full proof + pseudoinverse cases -i Supplement S2.

The partition becomes an imaging result only if one ordinary bucket stream can populate both ledgers with calibrated accuracy.

3 Pilot-free scene-and-medium inference

$$y_i = g_i a_i^\top x + \varepsilon_i, \quad \mathbb{E}[a] = \mathbf{1}, \quad (5)$$

► TODO: Read Eq. (5): g_i is the per-frame scalar gain path (the medium), the gauge $\mathbb{E}[a] = 1$ fixes the scene/gain scale split. Source: R36 s3 item 1.

► TODO: One paragraph: the pattern-normalized residual exposes the temporal gain fluctuations while scene scale enters only as an offset (R36 s3). Source: materials map s4 (DLGI model).

► TODO: ONE sentence on joint alternating inference – the single licensed mention of the smoother: name it a “Kalman–RTS smoother” here and nowhere else (each of the two names used exactly once); recursions -i Supplement S3.

► TODO: ONE sentence on the simulation-calibrated interval construction – the single licensed “Whittle” and “bootstrap” mentions (≤ 1 each); machinery -i Supplement S8.

► TODO: Name the four comparison arms in one clause: pilot-free (DLGI), pilot-interleaved, oracle monitor, and plain linear; state the identical-ledger rule (same photons, exposures, multiset; only order + inference change) and the schedule set {paired, random, ordered}. Source: R36 s3.

We therefore test the two products under one byte-identical acquisition ledger, against both a pilot monitor and an oracle path measurement.

4 Two products under one acquisition ledger

► TODO: Install the R34 authorized claim as the section’s load-bearing sentence (frozen, register B4, verbatim): “DLGI extracts a competitive point estimate of medium correlation time and fluctuation strength from the same bucket record used for scene reconstruction, with no added acquisition and no detected scene penalty; model-calibrated t_c uncertainty remains under confirmation.” On C1–C7 PASS, open the section with the one-line headline instead (R36 s4) and restore “model-certified” + validated scalar-gain domain (register B4).

► TODO: Claim 1 – medium product competitive + certified. t_c and CV risk against pilot and oracle; calibrated coverage; interval width on the declared interior grid. Numbers: $\langle C: \text{RMSE ratios} \leq 1.5 \times \text{pilot} \rangle$, $\langle C: \text{coverage} \in [0.92, 0.98] \rangle$, $\langle C: \text{width ratio} \rangle$. Evidence figure: Fig. 3. Bars + acceptance language -i Supplement S8.

► TODO: Claim 2 – scene product not spent. Byte-identical acquisition; scene noninferiority $\langle C: \Delta \text{PSNR} \geq -0.2 \text{ dB vs plain} \rangle$

FIG. 2 PLACEHOLDER --- reciprocity geometry and schedule consequence. (1) the joint Fisher block $[[A, C], [C^\top, B]]$. (2) whitening to K and the shared singular-value loss spectrum κ_i for the two Schur complements. (3) paired/random/ordered schedules with the preregistered PREDICTED ordering and the MEASURED ordering. One sentence: “Reducing canonical confusion protects both products.”

Figure 2: **[Fig. 2 title.]** [Caption 120–160 words. The figure tests the theorem’s schedule prediction independently of reconstruction quality (R36 s5 item 2). No full proof / no matrix zoo.]

linear); and the absence of extra photons, exposures, pilots, or duration (register B4 wording: “no additional photons, exposures, or duration under the matched acquisition ledger”; R35 cut 8 – NOT “zero cost”). Evidence figure: Fig. 4.

► TODO: Claim 3 – theorem predicts schedule + boundary. Reciprocity/schedule ordering (paired best for both ledgers) (C:ordering match); and the fast-drift, slow-drift, and weak-fluctuation limits (C:edge map). Evidence: Fig. 2 (schedule) + Fig. 4 (domain).

A second product is an instrument only inside the regime in which its model, intervals, and acquisition ledger remain valid.

5 Scope and experimental transfer

► TODO: Para 1 – physical scope (one paragraph). Scalar per-frame gain; fixed scene; validated residual model; declared cadence/span/CV domain. The exact physical violation that kills the model: spatially varying transfer rather than scalar gain. Source: R36 s5; R35 s7.4.

► TODO: Para 2 – closest-prior boundary (one paragraph). DCS/DWS, pilot monitoring, blind calibration, and AO/radio telemetry own their broad ingredients; the residual claim is simultaneous scene + medium delivery from the same programmable GI record with a joint information certificate. Keep it to a paragraph – the ruthless fence lives in Supplement S7 (R36 s6). FORBIDDEN novelty “firsts” per register B3.

► TODO: Para 3 – bench transfer (one paragraph). Rotating diffuser; known programmable target; synchronized bucket record; optional monitor used ONLY as validation truth. This pre-empts “no experiment” (R36 s5, s5 pre-emption). Source: R36 s5; paper2 “immediate bench test” seed (materials map s2a).

Methods and provenance capsule

► TODO: Methods stub. Cover tersely: (i) the exact statistical model + the “simulation model” labeling of every optical schematic; (ii) “preregistered numerical campaign” wording, NOT “experiment”; (iii) the scalar-gain observation model + gauge + physical-parameter mapping; (iv) a POINTER to the estimator + calibration algorithms in Supplement S3/S8 (do NOT name the smoother or resampling scheme again – word caps spent in Sec. 3); (v) ONE pointer to the R19 analysis-correction record + immutable artifacts in Supplement S4; (vi) code/data availability. Source: R36 s1 Methods row; materials map s4d.

Code and data availability

All simulation code, preregistration documents, manifests, correction records, and evidence bundles are available at [repo URL — USER DECISION (gap G7)].

Acknowledgments

[funding / acknowledgments — USER DECISION (gap G7)]

FIG. 3 PLACEHOLDER --- medium product: precision + calibration. LEFT: pilot-free/pilot RMSE ratios for t_c and CV across the primary grid, with the $1.5\times$ line and an oracle reference. RIGHT: empirical 95% coverage and median interval-width ratio, with the frozen acceptance band. No cell table; small multiples / compact heat map only if readable at column width. All values (C: gap G2).

Figure 3: **[Fig. 3 title.]** [Caption 120–160 words. Pilot and oracle baselines embedded so the evidence is not self-referential (R36 s5 item 4).]

FIG. 4 PLACEHOLDER --- scene product + validity domain. LEFT: scene noninferiority vs the byte-identical plain-linear comparator, -0.2 dB line, with the resource-ledger audit embedded as a small annotation. RIGHT: the validated operating domain in $(\Delta/t_c, T/t_c, CV, \text{photon-SNR})$ with fast-drift, slow-drift, and weak-fluctuation FAILURE regions clearly marked. One representative scene pair inset only if it explains what the quantitative panel cannot. No gallery.

Figure 4: **[Fig. 4 title.]** [Caption 120–160 words. Model falsifiability is part of the claim: the figure shows where the scalar-gain model fails (R36 s5 item 5).]